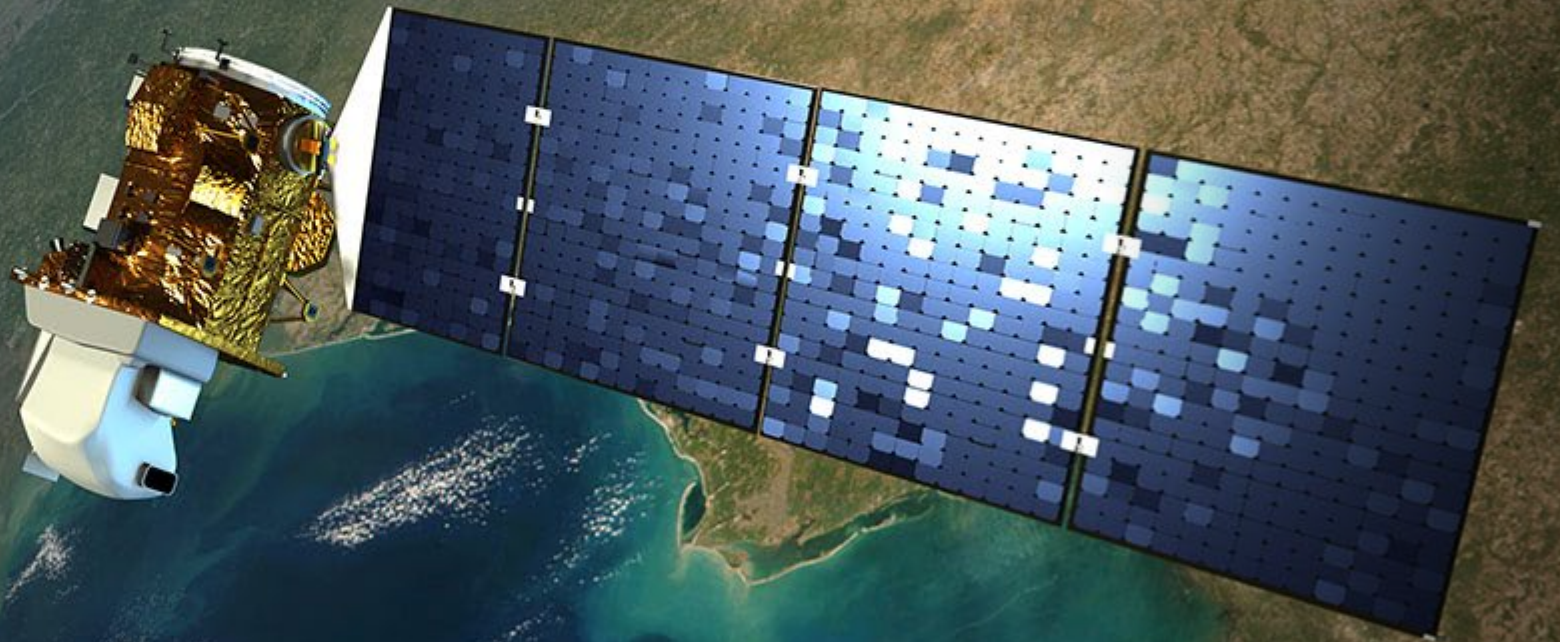
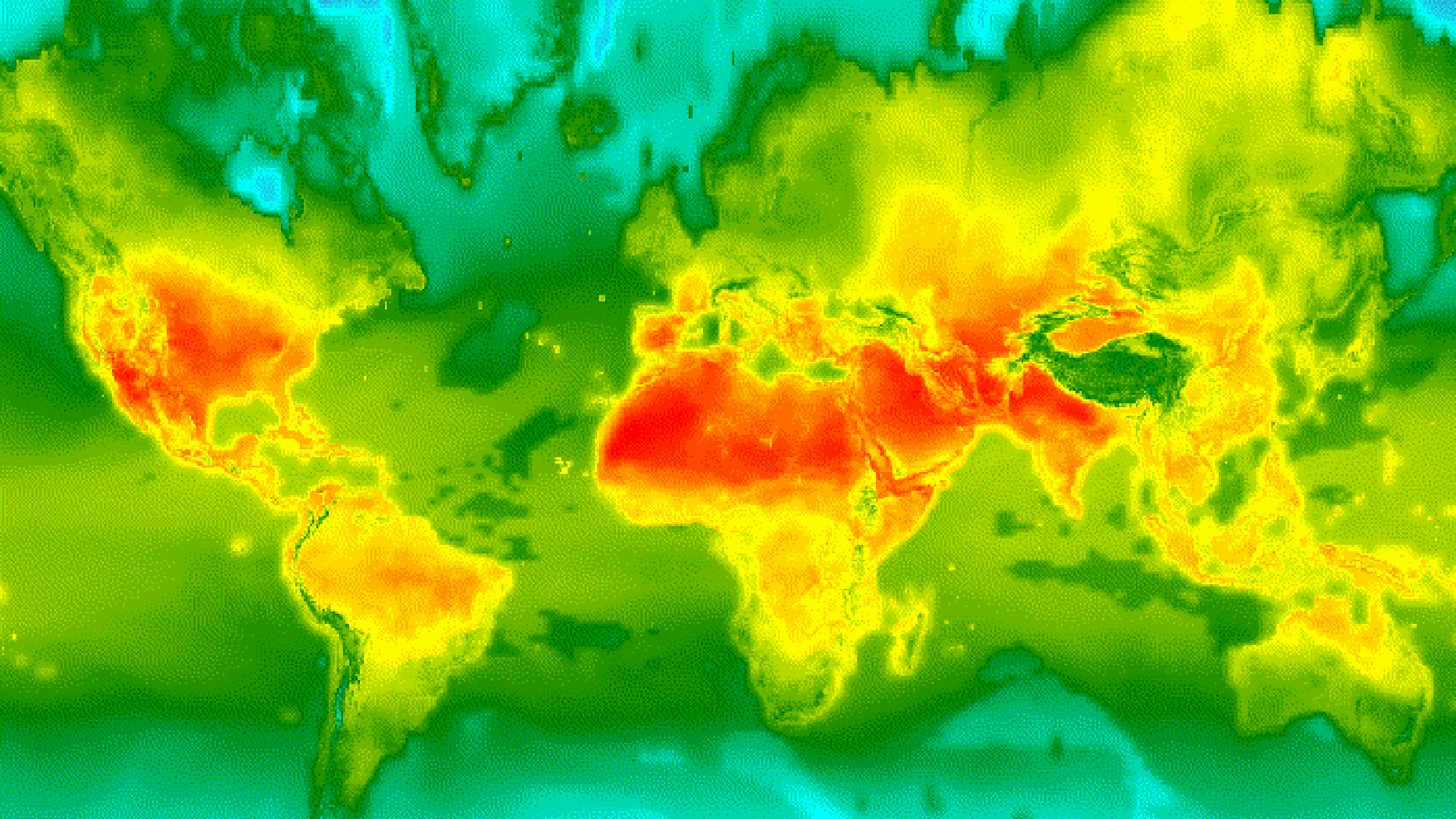
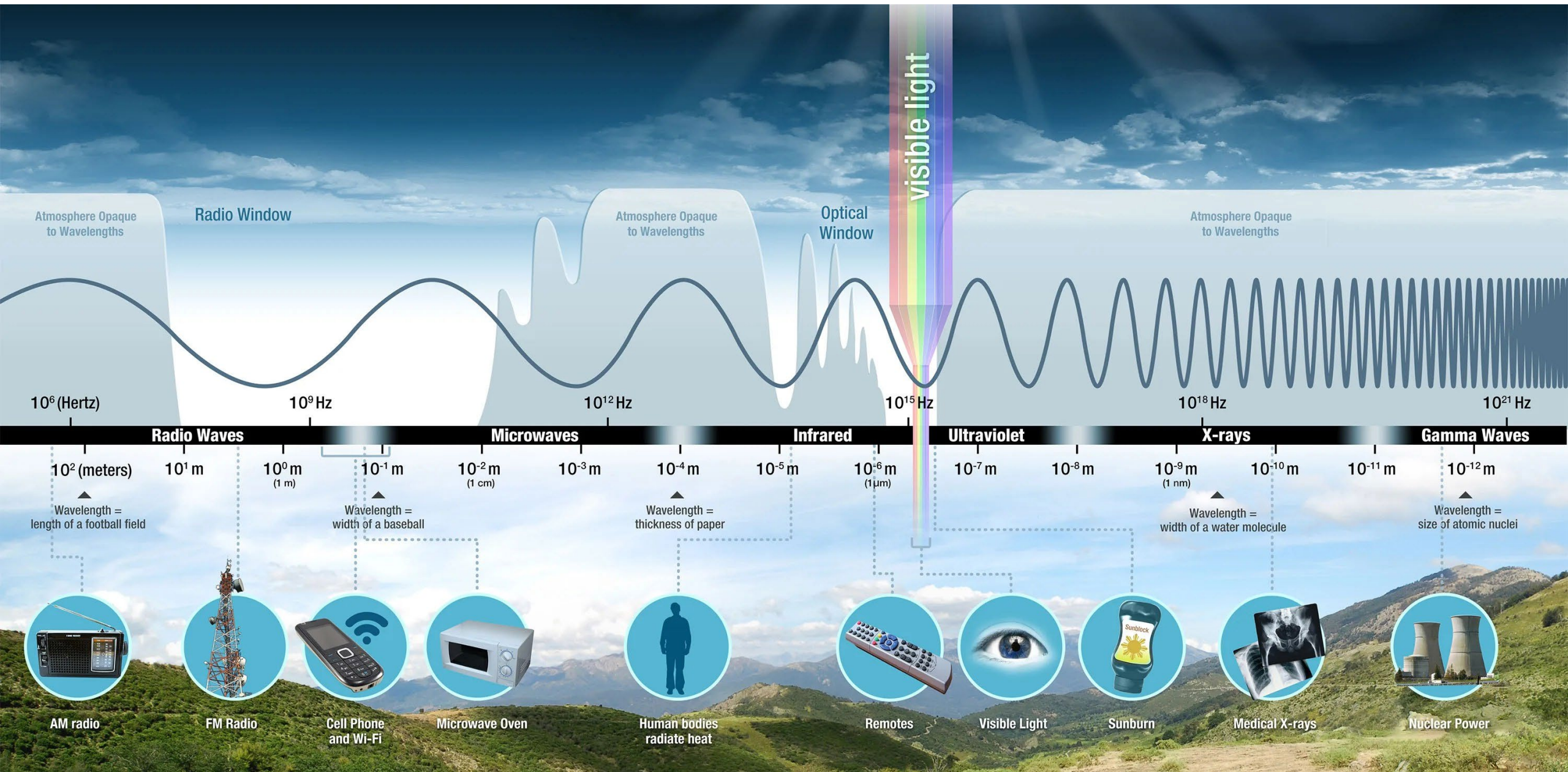


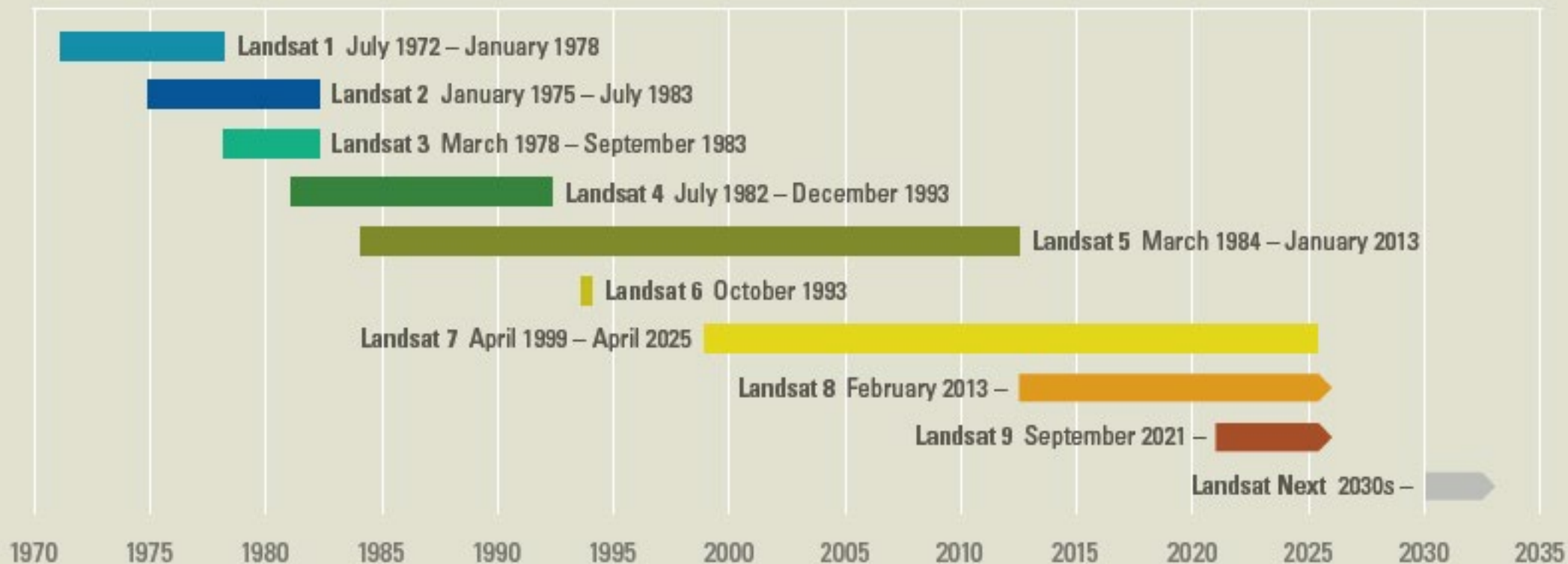
Remote Sensing Applications



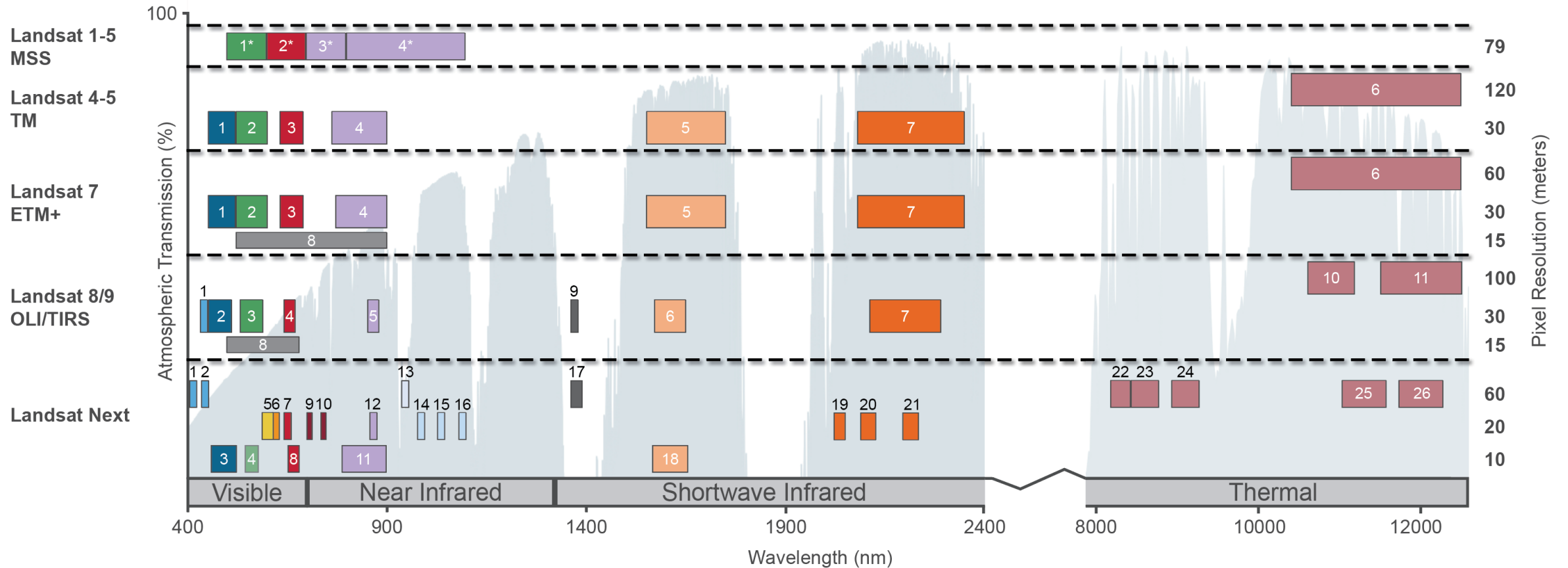




Landsat Missions: Imaging the Earth Since 1972



Comparison of Landsat Spectral Bands



* MSS bands 1–4 were known as bands 4–7, respectively, on Landsats 1–3

USGS, March 2024

NASA EARTH FLEET

OPERATING & FUTURE THROUGH 2023





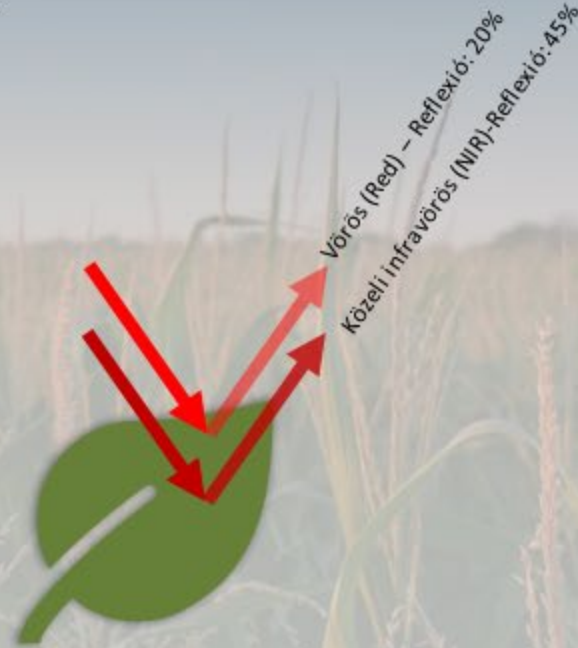
Calculation of NDVI

$$NDVI = \frac{NIR - RED}{NIR + RED}$$



Withered vegetation

$$NDVI = \frac{(0.40 - 0.35)}{(0.40 + 0.35)} = 0.07$$



Vegetation under stress

$$NDVI = \frac{(0.45 - 0.20)}{(0.45 + 0.20)} = 0.38$$



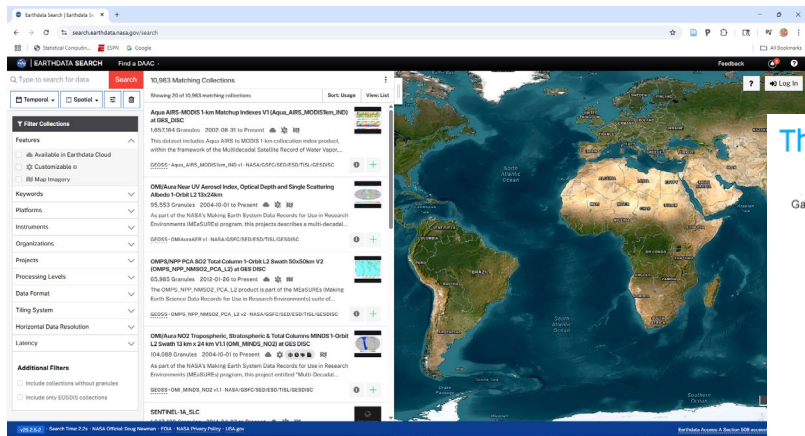
Healthy vegetation

$$NDVI = \frac{(0.60 - 0.09)}{(0.60 + 0.09)} = 0.74$$

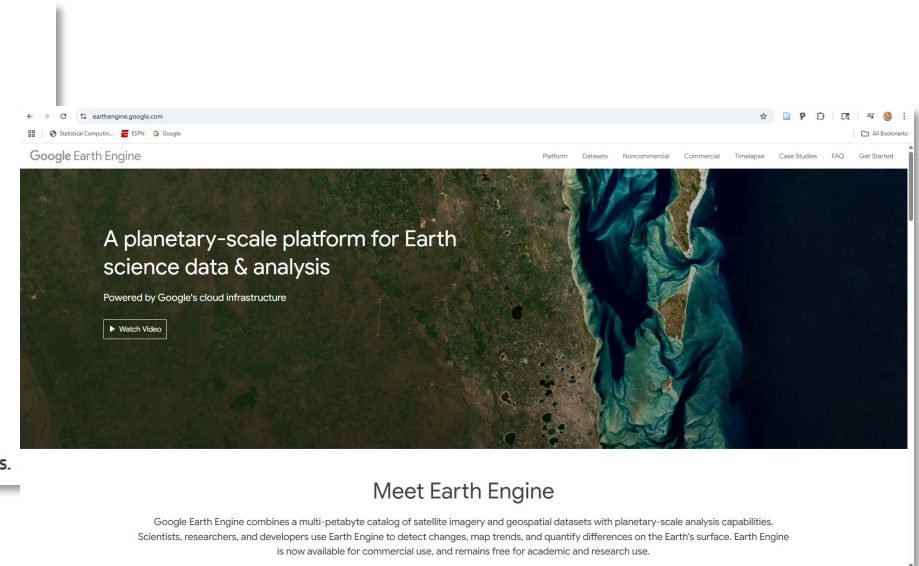
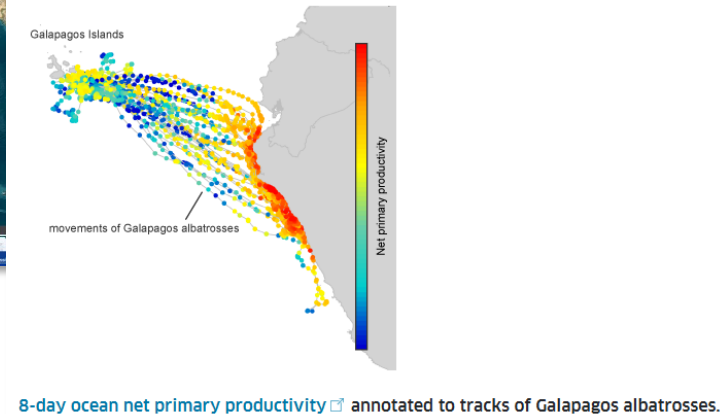
Where to Locate Satellite Data

Many sites exist to access data

- Navigate to:
 - NASA Earth Data - <https://search.earthdata.nasa.gov/search>
 - Movebank ENV Data Tool - <https://www.movebank.org/cms/movebank-content/env-data>
 - Google Earth Engine - <https://earthengine.google.com/>

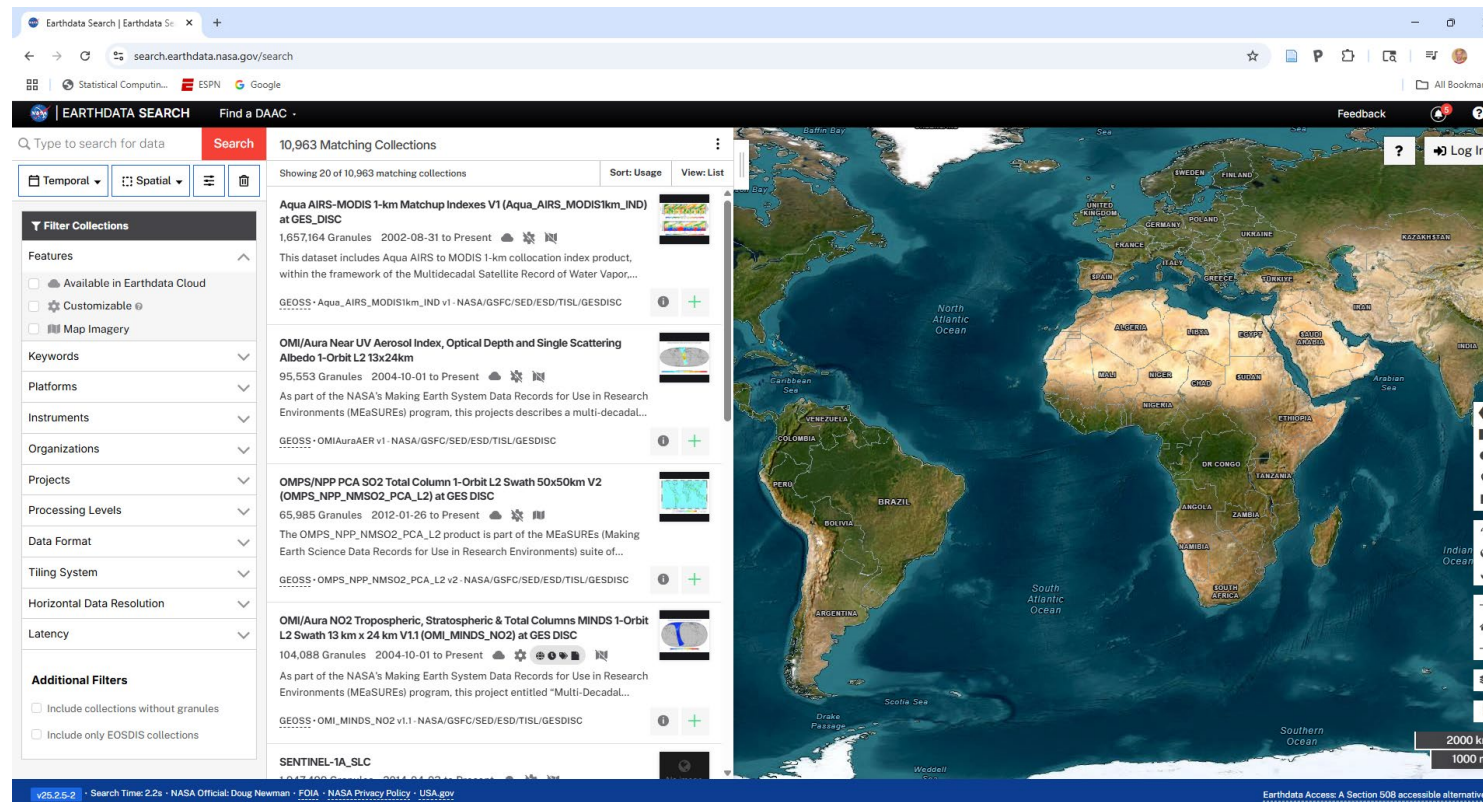


The Env-DATA System



NASA Earth Data

- Navigate to:
 - <https://search.earthdata.nasa.gov/search>



NASA Earth Data

- Enter Search Terms (vegetation, sea surface temperature, land cover, elevation, NDVI, ...)

The image displays the NASA Earth Data search interface. On the left, a search bar contains the text "NDVI" and a red "Search" button. Below the search bar are filters for "Temporal" and "Spatial", and a "Filter Collections" button. A red box highlights the search bar and filters. On the right, a list of search results is shown, including "MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061" and "MODIS/Terra Vegetation Indices Monthly L3 Global 1km SIN Grid V061". A red box highlights the "Filter Collections" button and the "Features" section. On the far right, a map of the world is shown, with a red box highlighting the "Filter Collections" button and the "Features" section.

NASA Earth Data

- Select product

MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061

170,637 Granules 2000-02-18 to Present

The Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices (MOD13Q1) Version 6.1 data are generated every 16 days a...

GEOSS • MOD13Q1 v061 - LP DAAC

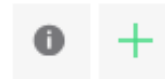


MODIS/Terra Vegetation Indices Monthly L3 Global 1km SIN Grid V061

88,939 Granules 2000-02-01 to Present

The Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices Monthly (MOD13A3) Version 6.1 data are provided monthl...

GEOSS • MOD13A3 v061 - LP DAAC

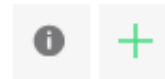


MODIS/Terra Vegetation Indices 16-Day L3 Global 500m SIN Grid V061

169,911 Granules 2000-02-18 to Present

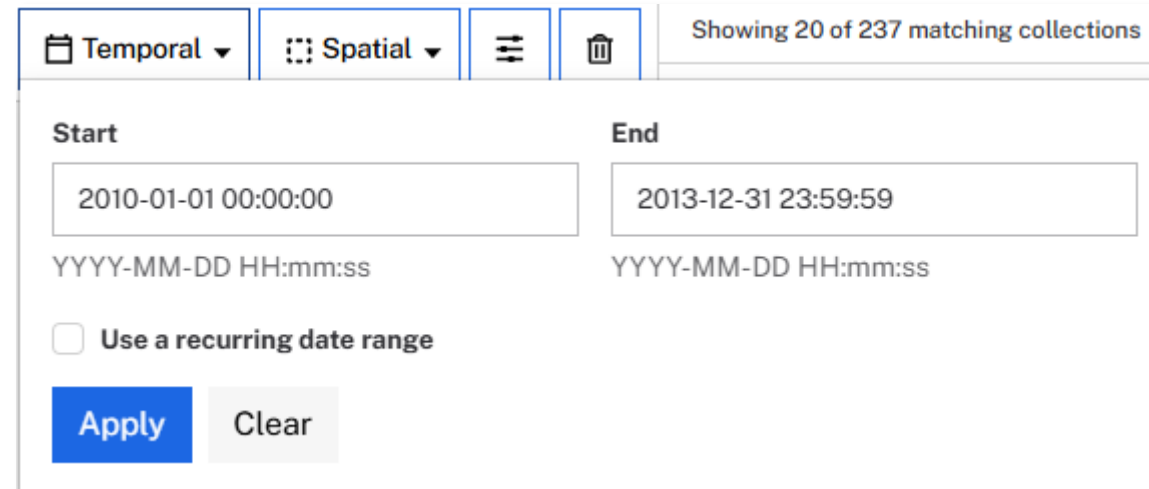
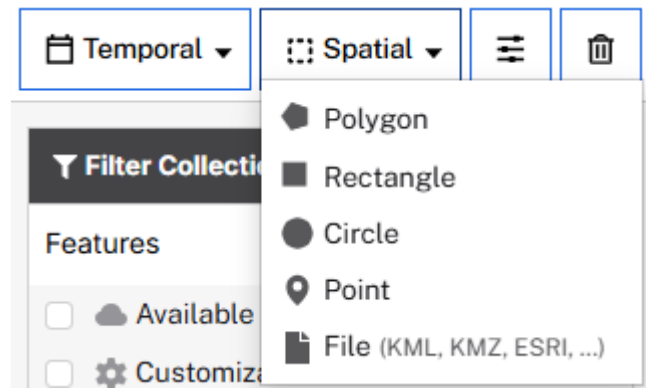
The MOD13A1 Version 6.1 product provides Vegetation Index (VI) values at a per pixel basis at 500 meter (m) spatial resolution. There are two primary...

GEOSS • MOD13A1 v061 - LP DAAC



NASA Earth Data

- Filter by spatial and temporal extent
Reduce the amount of data necessary to download



This screenshot shows the filter interface for NASA Earth Data. The 'Temporal' filter is selected, and the 'Spatial' filter is also visible. The 'Temporal' filter dropdown menu is open, showing options: Filter Collections, Features, Available, and Customized. The 'Spatial' filter dropdown menu is also open, showing options: Polygon, Rectangle, Circle, Point, and File (KML, KMZ, ESRI, ...). The 'Temporal' filter dropdown menu is also open, showing options: Filter Collections, Features, Available, and Customized.

Showing 20 of 237 matching collections

Start **End**

2010-01-01 00:00:00 2013-12-31 23:59:59

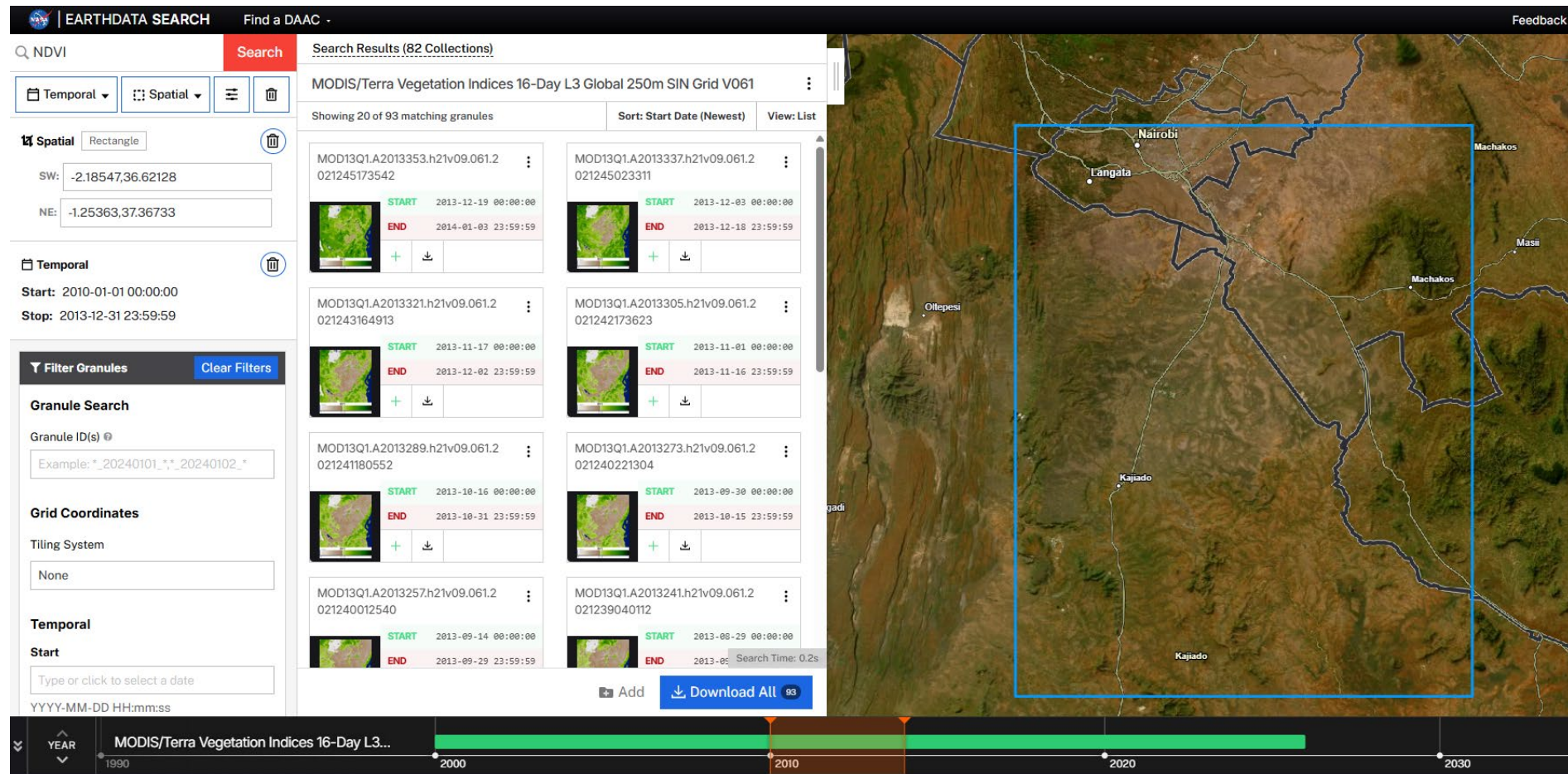
YYYY-MM-DD HH:mm:ss YYYY-MM-DD HH:mm:ss

☐ Use a recurring date range

Apply **Clear**

NASA Earth Data

- Check map and number of granules



NASA Earth Data

- Follow download instructions

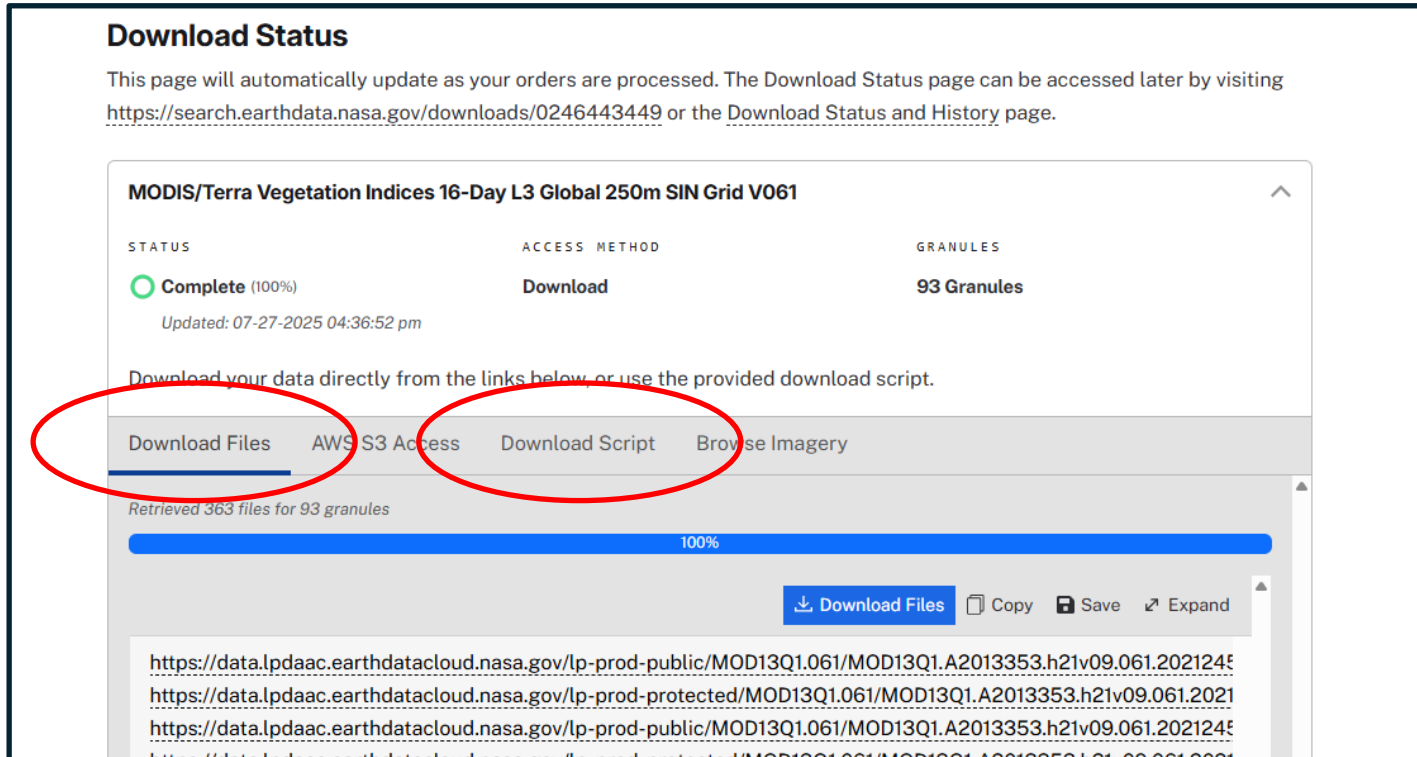
The screenshot displays the NASA Earth Data Search interface. On the left, a sidebar shows the project details: "Untitled Project" with 93 Granules, 1 Collection, and 20.0 GB. The selected dataset is "MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061". The main panel shows the "Edit Options" section with two steps: "1 Choose how you want to download your data" and "2 Select a service and customize options". Under step 1, the "Download all data" option is selected, with a sub-option "Direct download of all selected data". Step 2 indicates that no customization options are available for the selected access method. A "Download Data" button is at the bottom left. On the right, a satellite map of East Africa shows a blue rectangular selection box covering a region around Nairobi and Langata. The bottom of the interface features a timeline slider for the year, currently set to 2000, with markers for 1990, 2000, 2010, 2020, and 2030.

NASA Earth Data

- Follow download instructions

Option 1: Use the Earthdata Download tool (<https://nasa.github.io/earthdata-download/>)

Option 2: Use command-line tools (“Download Script” tab)



Download Status

This page will automatically update as your orders are processed. The Download Status page can be accessed later by visiting <https://search.earthdata.nasa.gov/downloads/0246443449> or the [Download Status and History](#) page.

MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061

STATUS	ACCESS METHOD	GRANULES
Complete (100%) Updated: 07-27-2025 04:36:52 pm	Download	93 Granules

Download your data directly from the links below, or use the provided download script.

Download Files | AWS S3 Access | **Download Script** | Browse Imagery

Retrieved 363 files for 93 granules

100%

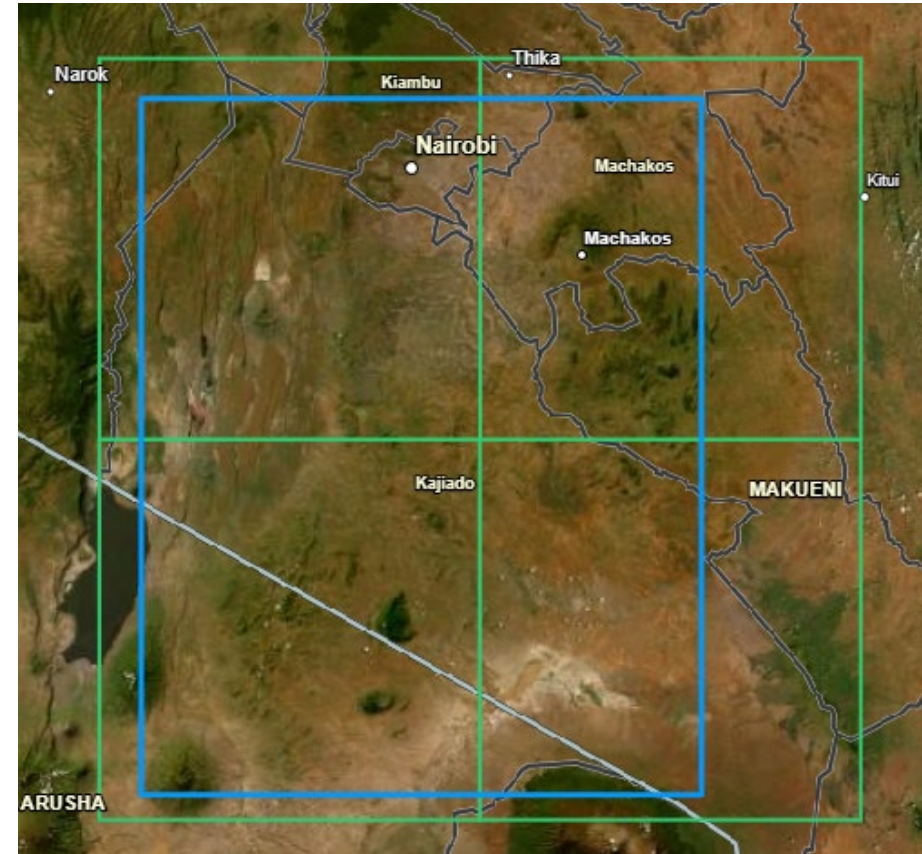
[Download Files](#) | [Copy](#) | [Save](#) | [Expand](#)

<https://data.lpdaac.earthdatacloud.nasa.gov/lp-prod-public/MOD13Q1.061/MOD13Q1.A2013353.h21v09.061.2021245>
<https://data.lpdaac.earthdatacloud.nasa.gov/lp-prod-protected/MOD13Q1.061/MOD13Q1.A2013353.h21v09.061.2021245>
<https://data.lpdaac.earthdatacloud.nasa.gov/lp-prod-public/MOD13Q1.061/MOD13Q1.A2013353.h21v09.061.2021245>

NASA Earth Data

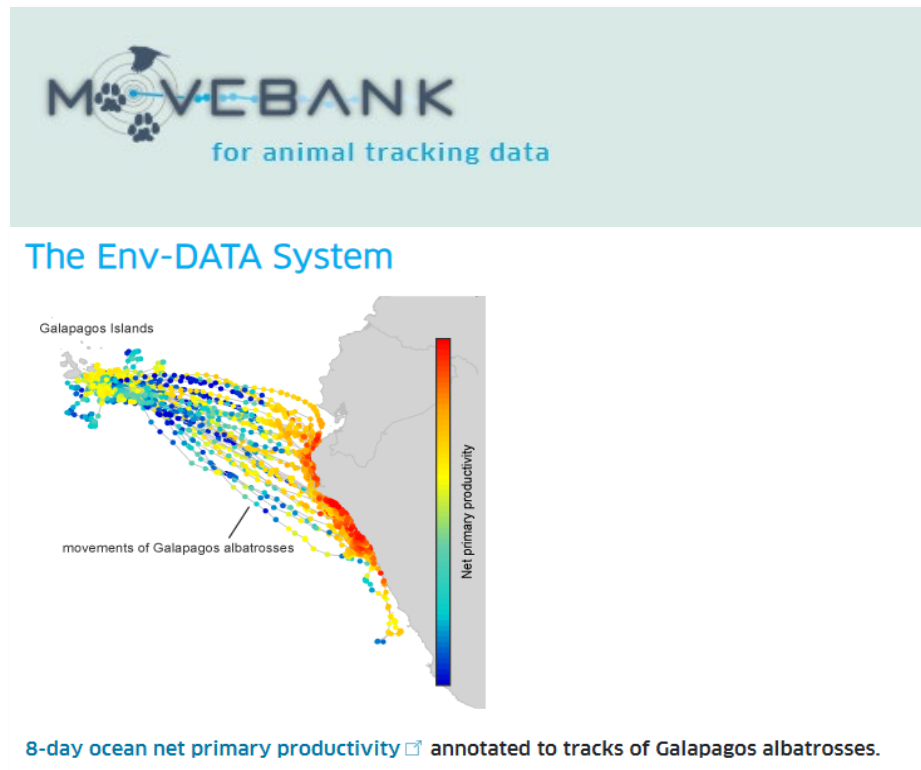
- Important Notes on Post-Processing
 - Many products will include multiple granules
 - E.g., Different locations or date/times
 - Many granules will include multiple layers
 - E.g., MODIS has multiple files for individual bands or projects (NDVI, EVI)
 - Requires GIS analyses
 - Mosaicking to combine spatial tiles
 - Stacking to combine temporal files
 - Cropping to include only need area

Example using SRTM elevation data: 4 tiles



Movebank ENV Data

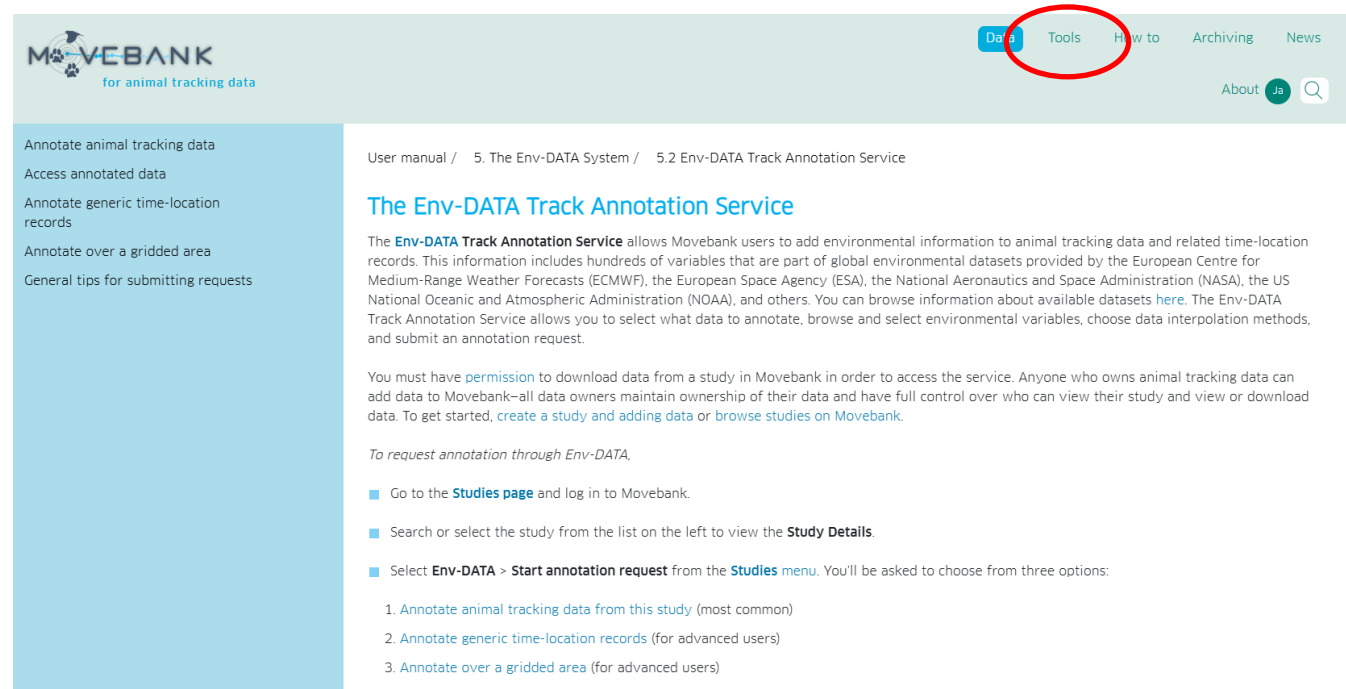
- Navigate to:
 - Movebank ENV Data Tool - <https://www.movebank.org/cms/movebank-content/env-data>



A set of free tools on Movebank to help you link animal movement data with information from global environmental datasets

Movebank ENV Data

- Go to Tools – EnvDATA:
 - Choose from:
 - Annotate animal tracking data from this study (most common) ←
 - Annotate generic time-location records (for advanced users)
 - Annotate over a gridded area (for advanced users)



Movebank ENV Data

- Start Annotation Request:
 - Go to **Study** and select **Start Annotation Request**

The screenshot displays the Movebank website for animal tracking data. The header features the Movebank logo and the tagline 'for animal tracking data'. Below the header, there is a navigation bar with a search bar and a 'Studies' tab. The 'Studies' tab is active, showing a list of studies on the left and details for the selected study on the right. The selected study is 'White-bearded wildebeest (Connochaetes taurinus) movements - Kenya'. The 'Start Annotation Request' button is circled in red in the top right corner of the 'Studies' tab.

Movebank
for animal tracking data

Search Studies

Studies where I am Data Manager

Filter by Study Name

Study - White-bearded wildebeest (Connochaetes taurinus) movements - Kenya
[Animals](#) [Tags](#) [Files](#) [Argos Feeds](#) [File Formats](#)

Study Details

Study Name White-bearded wildebeest (Connochaetes taurinus) movements - Kenya

Contact Person Jared

Principal Investigator Randall Boone

PI Contact Details 1499 Campus Delivery Natural Resource Ecology Laboratory
Colorado State University Fort Collins, CO 80523-1499
Stabach JA, Hughey L, Reid RS, Worden JS, Leimgruber P, Boone RB. 2020. Data from: Study "White-bearded wildebeest in Kenya". Movebank Data Repository. <https://www.doi.org/10.5441/001/1.h0t27719>

Citation Walton T, Boone R. 2017. Analyzing wildebeest behavior and effects of humans on activity in southwest Kenya. Poster [stored in this study; see Files]

Stabach JA. 2015. Movement, resource selection, and the physiological stress response of white-bearded wildebeest [dissertation]. [Fort Collins (CO, USA)]: Colorado State University. <https://doi.org/10.217/167207>

Boone RB, Reid RS, Lilieholm RJ, Worden JS, Ogutu JO. 2009. Wildebeest forage acquisition in fragmented landscapes under variable climates. National Science Foundation (NSF) DEB Grant 0810292

[Start Annotation Request](#)
[Show My Requests](#)
[About Env-DATA](#)

Movebank ENV Data

- Annotate animal tracking data

- Choose what animal tracking data you want to annotate. You can select from any of the Tags or Animals in the study, and can choose to include outliers and limit the annotation to one sensor type. When you are done select **Continue**.

Choose which data from this study you want to annotate

☒ Select animals

☐ Select tags

Available sensor types

GPS ▾

☐ Include outliers

Filter by date

Interval:

select interval ▾

From:

To:

☒ Nenkongu Narok

☒ Nentepesi

☒ Nesimej

☒ Nkairowua

☒ Nkoko

☒ Noontare

☒ Ntishya

☒ Ole Nagol

☒ Olope

☒ Paita

☒ Peria

☒ Reiyo

☒ Sawani

☒ Sotua

☒ Taletian

Select all

Deselect all

Cancel

Continue

Movebank ENV Data

- Annotate animal tracking data

- Next, browse and select the environmental variables you want to annotate. You can browse by [source](#)—such as MODIS Ocean or the ECMWF weather model—or by variable type—such as elevation or weather. To learn more about individual data sources or variables, click on ⓘ to view more information and links. **If you are requesting variables from multiple sources, we recommend splitting these into multiple smaller requests.**

Browse available environmental variables and select those you want to annotate

Browse available variables based on their source provider or the type of information they provide. If you will annotate many variables, we recommend submitting separate requests for each. The service is indicated in the first part of the name of selected variables.

[User manual](#)

Variables by sourceVariables by type

⊕ ASTER ASTGTM3 Global 30-m DEM ⓘ

⊕ ETOPO1 1-arc-minute Global Relief Model ⓘ

⊕ GlobCover

⊕ MODIS Land

⊕ MODIS Ocean

⊕ Movebank Derived Variables

⊕ NASA Distance to the Nearest Coast ⓘ

⊕ NASA Socioeconomic Data and Applications Center (SEDAC)

⊕ NCEP North American Regional Reanalysis (NARR)

⊕ NOAA Global Climate Indexes ⓘ

⊕ Oregon State Ocean Productivity Reanalysis

⊖ SRTM 1 Arc-Second DEM ⓘ

Variable

SRTM Elevation

Remove

CancelBackContinue

Derived from information provided by NASA's ARSET program – Introduction to the Integration of Animal Tracking and Remote Sensing

Movebank ENV Data

- Review and Submit (Send Annotation Request)
 - Data requests are available for download for **two** weeks as a .zip file
 - Annotated data will be shown as **available** in the **My requests** table (under **Env-Data** > **Show my requests** in the **Studies** menu).

Review a summary of your order and submit your annotation request

study:

White-bearded wildebeest
(Connochaetes taurinus)
movements - Kenya

sensor type:

GPS

tracking data:

Animal: Fifteen, Karbolo, Kayioni, Kikaya, Kiranto, Koyiaki, Laingoni, Ledama, Lempaash, Lengaruni, Leporana, Lerisa, Motorogi, Naboisho, Nagol, Naiba, Naisula, Naserian, Ne Mpaash, Neatoo, Nenkitirua, Nenkongu Narok, Nentepesi, Nesime, Nkairowua, Nkoko, Noontare, Ntishya, Ole Nagol, Olope, Paita, Peria, Reiyo, Sawani, Sotua, Taletian

Send download notification to:

jared_stabach@hotmail.com

Request name:

White-bearded wildebeest (Connochaetes t

Variable	Interpolation
SRTM Elevation	bilinear

Cancel

Back

Send annotation request

Note: Movebank requests can be time consuming. Best to split your data into smaller parts (i.e., create data subsets (a few animals) and request fewer environmental datasets (one or two) for annotation)

Movebank ENV Data

- Review and Submit (Send Annotation Request)
 - Data requests are available for download for **two** weeks as a .zip file
 - Annotated data will be shown as **available** in the **My requests** table (under **Env-Data** > **Show my requests** in the **Studies** menu).

MOVEBANK
for animal tracking data

Search Studies

Studies where I am Data Manager New Study

Filter by Study Name

Study - White-bearded wildebeest (Connochaetes taurinus) movements - Kenya
Animals Tags Files Arcs Feeds File Formats

Study Details

Study Name White-bearded wildebeest (Connochaetes taurinus) movements - Kenya

Contact Person Jared

Principal Investigator Randall Boone

PI Contact Details 1499 Campus Delivery Natural Resource Ecology Laboratory Colorado State University Fort Collins, CO 80523-1499 Stabach JA, Hughey L, Reid RS, Worden JS, Leimgruber P, Boone RB. 2020. Data from: Study "White-bearded wildebeest in Kenya". Movebank Data Repository <https://www.doi.org/10.5441/001/1.h0127719>

Citation Walton T, Boone R. 2017. Analyzing wildebeest behavior and effects of humans on activity in southwest Kenya. Poster [stored in this study; see Files]

Stabach JA. 2015. Movement, resource selection, and the physiological stress response of white-bearded wildebeest [dissertation]. [Fort Collins (CO, USA)]: Colorado State University. <https://doi.org/10.1021/167207>

Boone RB, Reid RS, Lilieholm RJ, Worden JS, Ogutu JO. 2009. Wildebeest forage acquisition in fragmented landscapes under variable climates. National Science Foundation (NSF) DEB

My requests

[User manual](#)

Find a list of your submitted annotation requests below. Click on the links in the table for information on submitted requests. Once results are available, download annotated data from the link on the right side of the table. Processing time depends on the spatial and temporal scope and resolution of your dataset and the requested environmental attributes, the number of other requests being processed and whether others have recently requested the same environmental data. If your request fails or you have not gotten your results within a few weeks, please send the access key to support@movebank.org for assistance. You can copy the access key from under **Details** (third paragraph) below. [Read more](#) for guidance on interpreting your results and answers to frequently asked questions.

Request	Timestamp	State	Overview	Details
White-bearded wildebeest (Connochaetes taurinus) movements - Kenya	2025-07-28 12:13:52.670	available	view	view download

[Close](#) [Refresh](#)

Note: Movebank requests can be time consuming. Best to split your data into smaller parts (i.e., create data subsets (a few animals) and request fewer environmental datasets (one or two) for annotation)

Movebank ENV Data – Other Applications

- Annotate generic time-location records

Get environmental information for times and locations related to but not the same as the animal location themselves (e.g., simulated trajectories or randomly sampled locations)

Upload data for annotation

Here you can request annotation of generic time-location records related to your tracking data on Movebank. Select from a list of previously submitted files or upload a new file.

Uploaded files must be comma separated with a header including the following fields:

- timestamp (required, date and time in UTC in this format: yyyy-MM-dd HH:mm:ss.SSS)
- location-lat (required, longitude projected in WGS84 in decimal degrees, range -180 to 180)
- location-long (required, latitude projected in WGS84 in decimal degrees)
- height-above-msl (optional, height above mean sea level in meters)

Other fields are allowed but will not be evaluated for annotation. As noted above, all calculations will assume that times are in UTC, location coordinates are projected in WGS84, and heights represent height above mean sea level. Height above ellipsoid needs to be converted before submission. Null values should be left blank. For additional information see [here](#)

[Download example data](#)

Upload file

Choose File

No file chosen

Cancel

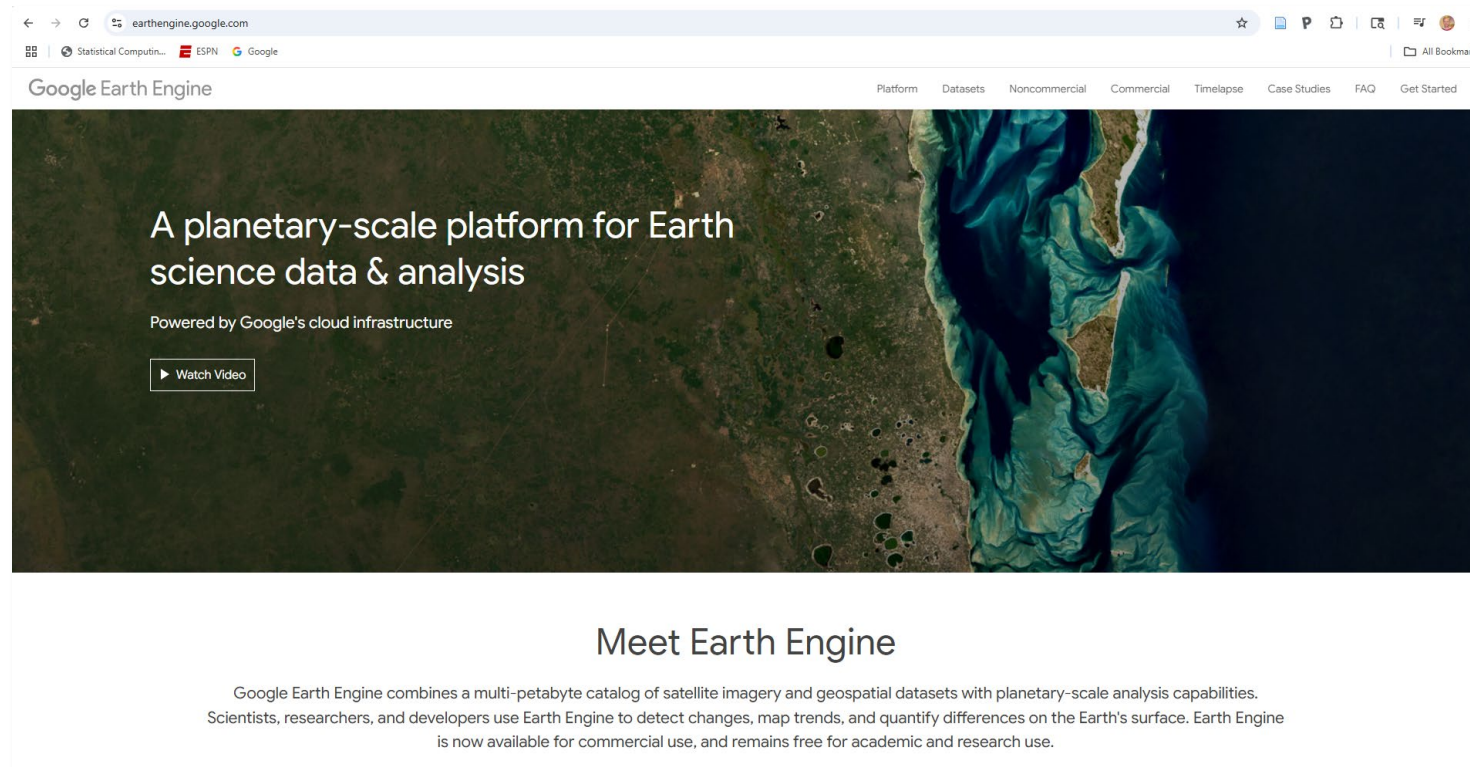
Continue

Note: Make sure the uploaded files matches the required format **EXACTLY**

Google Earth Engine

A Multi-petabyte catalog and geospatial datasets with planetary-scale analysis capabilities

- Navigate to:
 - Google Earth Engine - <https://earthengine.google.com/>



Google Earth Engine

- Follow GEE workflow

Open Access Communication

Enhancing Animal Movement Analyses: Spatiotemporal Matching of Animal Positions with Remotely Sensed Data Using Google Earth Engine and R

by Ramiro D. Crego ^{1,2,*} , Majaliwa M. Masolele ³ , Grant Connette ^{1,2}  and Jared A. Stabach ¹ 

